THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5	Distance formula
2	**(c)** $(2,\ 3)$	Section formula
3	**(c)** 12	$\frac{1}{2}\times$ base $\times$ height
4	**(d)** 8	$x = 2m - a$
5	**(a)** 5	Distance from y-axis $= \lvert x\rvert$
6	**(b)** 3	Equate the squared distances
7	**(b)** 13	Distance formula
8	**(b)** $(5,\ 6)$	Section formula
9	**(b)** 20	$\frac{1}{2}\times$ base $\times$ height
10	**(b)** 10	$x = 2m - a$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ (1); $= \sqrt{(2)^2+(-2)^2} = \sqrt{8}$ (1); $= \sqrt{8}$ units (1).	1+1+1
12	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{3(8)+1(4)}{4} = 7$ (1); $y = 3$ (1).	1+1+1
13	$(8)^2 + (y+3)^2 = 100$ (1); $(y+3)^2 = 36$ (1); $y = -3 \pm 6$, i.e. $y = 3$ or -9 (1).	1+1+1
14	$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ (1); $= \sqrt{(4)^2+(-4)^2} = \sqrt{32}$ (1); $= \sqrt{32}$ units (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of triangle must be 0 (1); $\frac{1}{2}\lvert 1(6-k)+3(k-2)+5(2-6)\rvert = 0$ (1); $k = 10$ (1).	1+1+1
16	Area $= \frac{1}{2}\lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert 3+4-28\rvert$ (1); $= \frac{21}{2}$ sq. units (1).	1+1+1
17	Let ratio $m : n$; $y = \frac{m(8)+n(-6)}{m+n} = 0$ (2); $m : n = \frac{3}{4} : 1$, i.e. 3:4 (1); point $x = \frac{3(3)+4(-4)}{7} = -1$ (2).	2+1+2
	OR: Area of triangle must be 0 (1); $\frac{1}{2}\lvert 2(5-k)+4(k-1)+6(1-5)\rvert = 0$ (1); $k = 9$ (1).	1+1+1
18	Area $= \frac{1}{2}\lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert 8+7+6\rvert$ (1); $= \frac{21}{2}$ sq. units (1).	1+1+1
	OR: Let ratio $m : n$; $y = \frac{m(9)+n(-3)}{m+n} = 0$ (2); $m : n = \frac{1}{3} : 1$, i.e. 1:3 (1); point $x = \frac{1(6)+3(2)}{4} = 3$ (2).	2+1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let the point be $(x, 0)$; $(x-7)^2+36 = (x+3)^2+16$ (2); $x = 3$ (2); point $(3, 0)$ (1).	2+2+1
	OR: Diagonals bisect each other: midpoint of AC is $\left(3, \frac{9}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (2,\ 6)$ (2).	2+1+2
20	Let the point be $(x, 0)$; $(x-2)^2+9 = (x-8)^2+1$ (2); $x = \frac{13}{3}$ (2); point $(\frac{13}{3}, 0)$ (1).	2+2+1
	OR: Diagonals bisect each other: midpoint of AC is $\left(\frac{9}{2}, \frac{7}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (3,\ 5)$ (2).	2+1+2

Mistakes to avoid:

- Using $(x_1 - x_2)$ and $(y_2 - y_1)$ inconsistently or forgetting to square a negative difference.
- Leaving the square root out, giving AB^2 instead of AB.
- In the section formula, swapping the ratio: the multiplier of x_2 is m_1 (not m_2).
- Using the mid-point formula with differences instead of sums.
- Writing a point on the x-axis as $(0, y)$; it is $(x, 0)$.
- Declaring a triangle right-angled without checking that the *longest* side squared equals the sum of the other two.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 7	Distance from y-axis $= \lvert x \rvert$
2	**(a)** $\frac{3}{2}$	Equate the squared distances
3	**(b)** 15	Distance formula
4	**(d)** $(0,\ 2)$	Section formula
5	**(a)** 15	$\frac{1}{2}\times$ base $\times$ height
6	**(a)** 8	$x = 2m - a$
7	**(b)** 9	Distance from y-axis $= \lvert x \rvert$
8	**(b)** $\frac{7}{2}$	Equate the squared distances
9	**(c)** 10	Distance formula
10	**(b)** $(3,\ 5)$	Section formula

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{2(4)+3(-1)}{5} = 1$ (1); $y = 3$ (1).	1+1+1
12	$(6)^2 + (y-1)^2 = 100$ (1); $(y-1)^2 = 64$ (1); $y = 1 \pm 8$, i.e. $y = 9$ or -7 (1).	1+1+1
13	$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$ (1); $= \sqrt{(-5)^2 + (12)^2} = \sqrt{169}$ (1); $= 13$ units (1).	1+1+1
14	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{1(10)+3(2)}{4} = 4$ (1); $y = 4$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of triangle must be 0 (1); $\frac{1}{2}\lvert -1(7-k) + 1(k-3) + 3(3-7)\rvert = 0$ (1); $k = 11$ (1).	1+1+1
16	Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert 0 - 15 - 12\rvert$ (1); $= \frac{27}{2}$ sq. units (1).	1+1+1
17	Let ratio $m : n$; $y = \frac{m(8)+n(-4)}{m+n} = 0$ (2); $m : n = \frac{1}{2} : 1$, i.e. 1:2 (1); point $x = \frac{1(6)+2(0)}{3} = 2$ (2).	2+1+2
	OR: Area of triangle must be 0 (1); $\frac{1}{2}\lvert 0(7-k) + 3(k-1) + 4(1-7)\rvert = 0$ (1); $k = 9$ (1).	1+1+1
18	Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert -24 - 12 - 4\rvert$ (1); $= 20$ sq. units (1).	1+1+1
	OR: Let ratio $m : n$; $y = \frac{m(4)+n(-2)}{m+n} = 0$ (2); $m : n = \frac{1}{2} : 1$, i.e. 1:2 (1); point $x = \frac{1(4)+2(-2)}{3} = 0$ (2).	2+1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let the point be $(x, 0)$; $(x+1)^2 + 25 = (x-5)^2 + 9$ (2); $x = \frac{2}{3}$ (2); point $(\frac{2}{3}, 0)$ (1).	2+2+1
	OR: Diagonals bisect each other: midpoint of AC is $\left(\frac{3}{2}, \frac{5}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (0,\ 4)$ (2).	2+1+2
20	Let the point be $(x, 0)$; $(x-4)^2 + 4 = (x+2)^2 + 36$ (2); $x = -\frac{5}{3}$ (2); point $(-\frac{5}{3}, 0)$ (1).	2+2+1
	OR: Diagonals bisect each other: midpoint of AC is $(3, 6)$ (2); this is also the midpoint of BD (1); $D = (1,\ 8)$ (2).	2+1+2

Mistakes to avoid:

- Using $(x_1 - x_2)$ and $(y_2 - y_1)$ inconsistently or forgetting to square a negative difference.
- Leaving the square root out, giving AB^2 instead of AB.
- In the section formula, swapping the ratio: the multiplier of x_2 is m_1 (not m_2).
- Using the mid-point formula with differences instead of sums.
- Writing a point on the x-axis as $(0, y)$; it is $(x, 0)$.
- Declaring a triangle right-angled without checking that the *longest* side squared equals the sum of the other two.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 21	$\frac{1}{2}\times$ base $\times$ height
2	**(b)** -4	$x = 2m - a$
3	**(c)** 6	Distance from y-axis $= \lvert x\rvert$
4	**(a)** 7	Equate the squared distances
5	**(a)** 13	Distance formula
6	**(b)** $(3,\ 4)$	Section formula
7	**(a)** 18	$\frac{1}{2}\times$ base $\times$ height
8	**(d)** 13	$x = 2m - a$
9	**(c)** 8	Distance from y-axis $= \lvert x\rvert$
10	**(b)** $\frac{11}{3}$	Equate the squared distances

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(6)^2 + (y-2)^2 = 100$ (1); $(y-2)^2 = 64$ (1); $y = 2 \pm 8$, i.e. $y = 10$ or -6 (1).	1+1+1
12	$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$ (1); $= \sqrt{(12)^2 + (-5)^2} = \sqrt{169}$ (1); $= 13$ units (1).	1+1+1
13	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{2(6)+3(-4)}{5} = 0$ (1); $y = 2$ (1).	1+1+1
14	$(8)^2 + (y-5)^2 = 100$ (1); $(y-5)^2 = 36$ (1); $y = 5 \pm 6$, i.e. $y = 11$ or -1 (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of triangle must be 0 (1); $\frac{1}{2}\lvert 1(5-k) + 3(k+1) + 4(-1-5)\rvert = 0$ (1); $k = 8$ (1).	1+1+1
16	Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert 4 - 32 + 3\rvert$ (1); $= \frac{25}{2}$ sq. units (1).	1+1+1
17	Let ratio $m : n$; $y = \frac{m(10)+n(-5)}{m+n} = 0$ (2); $m : n = \frac{1}{2} : 1$, i.e. 1:2 (1); point $x = \frac{1(7)+2(1)}{3} = 3$ (2).	2+1+2
	OR: Area of triangle must be 0 (1); $\frac{1}{2}\lvert 2(7-k) + 6(k-3) + 8(3-7)\rvert = 0$ (1); $k = 9$ (1).	1+1+1
18	Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert 4 + 10 - 42\rvert$ (1); $= 14$ sq. units (1).	1+1+1
	OR: Let ratio $m : n$; $y = \frac{m(6)+n(-6)}{m+n} = 0$ (2); $m : n = 1 : 1$, i.e. 1:1 (1); point $x = \frac{1(9)+1(3)}{2} = 6$ (2).	2+1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let the point be $(x, 0)$; $(x-1)^2 + 49 = (x-9)^2 + 1$ (2); $x = 2$ (2); point $(2, 0)$ (1).	2+2+1
	OR: Diagonals bisect each other: midpoint of AC is $(5, 7)$ (2); this is also the midpoint of BD (1); $D = (3,\ 9)$ (2).	2+1+2
20	Let the point be $(x, 0)$; $(x-3)^2 + 4 = (x-7)^2 + 64$ (2); $x = \frac{25}{2}$ (2); point $(\frac{25}{2}, 0)$ (1).	2+2+1
	OR: Diagonals bisect each other: midpoint of AC is $\left(\frac{7}{2}, \frac{7}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (2,\ 5)$ (2).	2+1+2

Mistakes to avoid:

- Using $(x_1 - x_2)$ and $(y_2 - y_1)$ inconsistently or forgetting to square a negative difference.
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